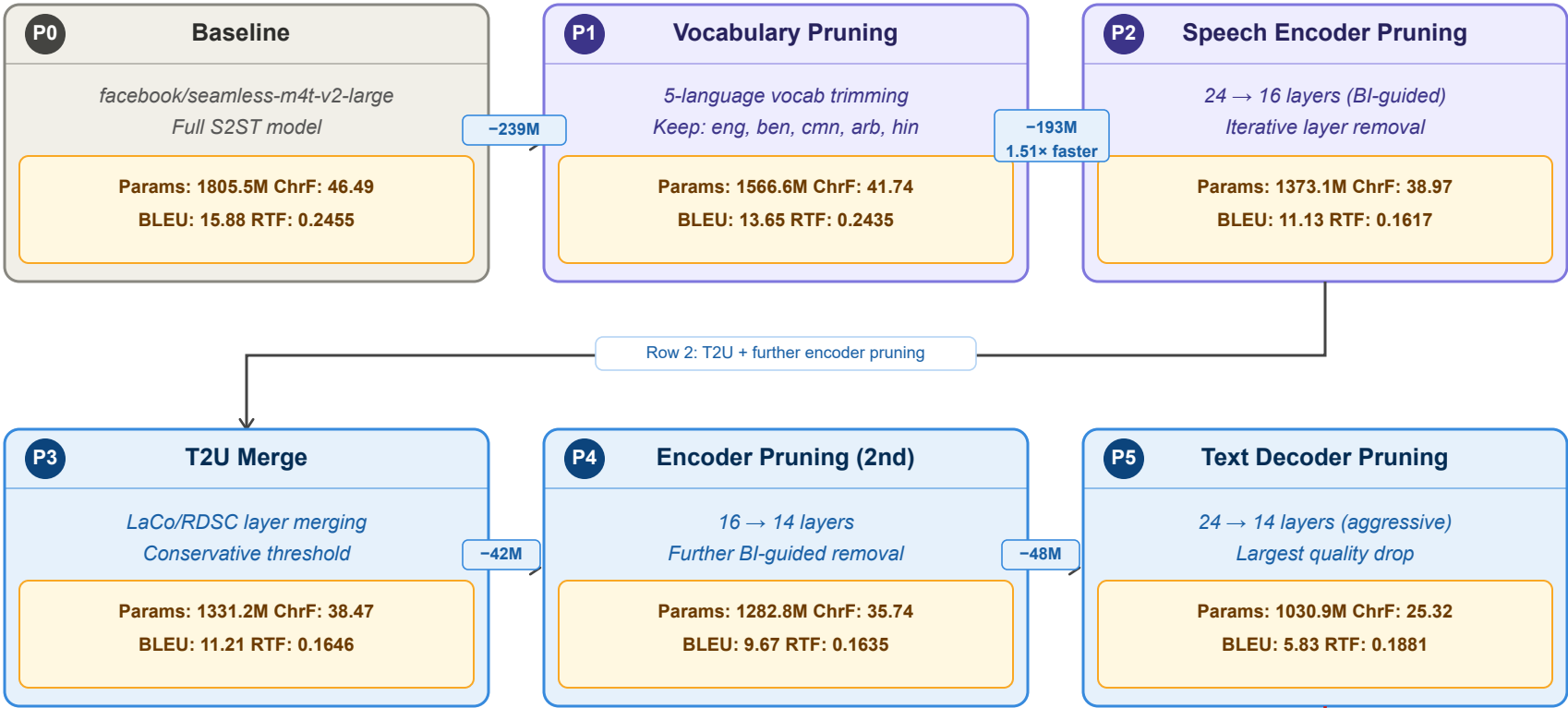


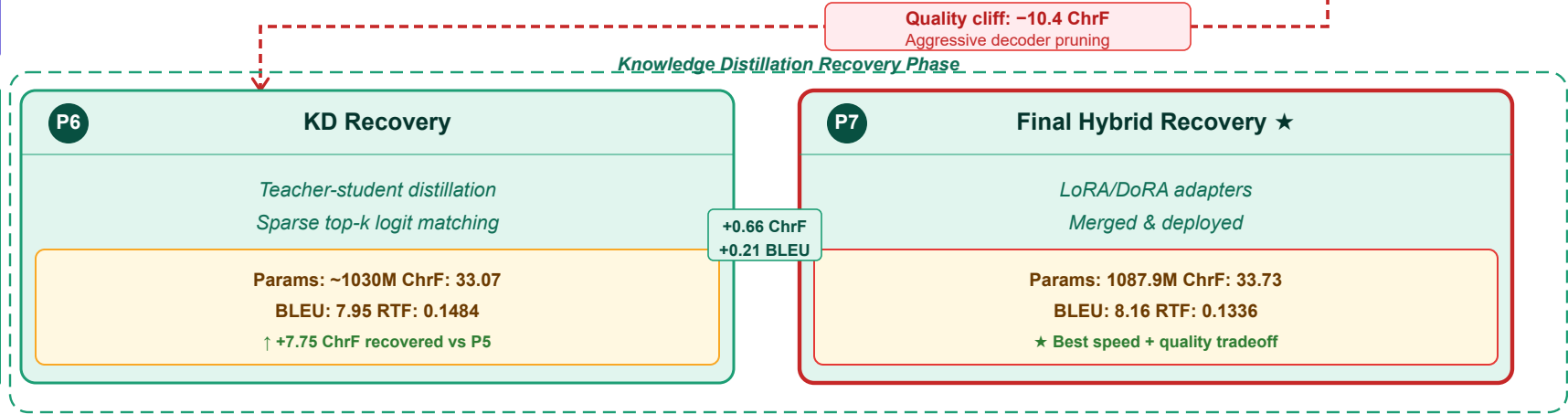
SeamlessM4Tv2 Compression Pipeline: Pruning & Recovery

Structured Pruning with Knowledge Distillation Recovery

STRUCTURAL PRUNING



RECOVERY



Compression Summary

- 39.7% parameter reduction
- 1805M → 1088M params
- 1.84× RTF speedup
- 72.6% ChrF retained
- Runs on 4 GB GPU
- 8 language pairs tested

Key Techniques

- Block Influence (BI) scoring
- Iterative layer removal
- Conservative T2U merge
- Sparse KD recovery
- LoRA / DoRA adapters

Failed Attempts

- ✗ FLAP width pruning (collapsed generation)
- ✗ Sub-500M param target (destroyed quality)
- Defined safe boundary

Legend: Baseline Vocab / Encoder pruning T2U / Decoder pruning KD Recovery Final model ★

8 language pairs: eng↔ben, eng↔cmn, eng↔arb, eng↔hin | Evaluation: 25 samples/pair on FLEURS